

Simon Marais 2025 — B1

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A function $f : \{1, 2, \dots, n\} \rightarrow \{m+1, m+2, \dots, m+n\}$, where m and n are positive integers, is said to be *beaut* if

- a divides $f(a)$ for all $a \in \{1, 2, \dots, n\}$, and
- $f(a) \neq f(b)$ for all distinct elements a and b of $\{1, 2, \dots, n\}$ with $\gcd(a, b) \neq 1$.

Determine all positive integers n such that, for every positive integer m , there exists a beaut function $f : \{1, 2, \dots, n\} \rightarrow \{m+1, m+2, \dots, m+n\}$.

Solution

First we are going to show that for $n \geq 6$, a beaut function never exists.

Suppose n is even and $n \geq 6$. Pick $m = n(n-2) - 3$ so that the image of f is

$$\{n(n-2) - 2, n(n-2) - 1, n(n-2), \dots, n(n-2) + (n-3)\}$$

If we reduce this set modulo n or $(n-2)$ it becomes

$$\{-2, -1, 0, 1, 2, \dots, (n-3)\}$$

So $n(n-2)$ is the only number that is divisible by n or $(n-2)$. Now if a beaut function f existed we would require $f(n) = f(n-2) = n(n-2)$ due to the divisibility condition of beaut functions. However $\gcd(n, n-2)$ is a multiple of 2 since both $n, (n-2)$ are even. But this contradicts the second condition of beaut functions.

Similarly suppose n is odd and $n \geq 7$. Pick $m = (n-1)(n-3) - 4$ so that the image of f is

$$\{(n-1)(n-3) - 3, (n-1)(n-3) - 2, (n-1)(n-3) - 1, (n-1)(n-3), \dots, (n-1)(n-3) + (n-4)\}$$

Reducing this set modulo $(n-1)$ or $(n-3)$ we have

$$\{-3, -2, -1, 0, 1, \dots, (n-4)\}$$

So $(n-1)(n-3)$ is the only number divisible by $(n-1)$ or $(n-3)$. If a beaut function f existed we would need $f(n-1) = f(n-3) = (n-1)(n-3)$ due to the divisibility condition of beaut functions. However $\gcd(n-1, n-3)$ is even since both $(n-1), (n-3)$ are even. This contradicts the second condition of beaut functions.

It remains to show that beaut functions do exist for $n = 1, 2, 3, 4, 5$. The $n = 1$ case is trivial since 1 divides every integer. For the remaining n , note that any set of n consecutive numbers will have all the residues mod n , so we will be able to find numbers in the image of f such that $a \in \{1, 2, \dots, n\}$ divides $f(a)$. Since 2, 4 is the only pair of non-coprime integers here we just need to ensure $f(2) \neq f(4)$. This is possible since in any set of ≥ 4 numbers there should be one number $\equiv 0 \pmod{4}$ and one number $\equiv 2 \pmod{4}$. ■